

Villanova University

Department of Mechanical Engineering

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**FINAL REPORT 2. THERMAL FLUID DESIGN OF A SERVER LIQUID
COOLING SYSTEM**

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1. Comparison of Parallel Fins to Pin Fins in Cold Plate Design

This final report compared two cold plate designs for high-powered processing units in a server rack. The main factors considered were the effectiveness of the plates, as determined by their thermal resistance, and the pressure drop through the plates. Additionally, this report designed the flow loop to deliver cooling water to all eight chips on a single rack. This design considered the necessary piping and components, as well as the pressure drop through each fitting, and the required flow rate to maintain an effective chip temperature.

1.1 Revision of Interim Report Parallel Fin Cold Plate Model

The preliminary cold plate model was validated in the Interim Report. All experimental data provided for a cold plate of the listed geometry matched the simulation results for those same geometrical inputs. The only discrepancy between the experimental data and the preliminary model was the predicted pressure drop. The preliminary model calculated a pressure drop three times that of the measured pressure drop data provided for the same fin geometry.

Upon observation of this discrepancy, all equations used in the preliminary model were checked for errors, and no noticeable mistakes were found. All mathematical formulas were accurate and verified with equations from vetted literature. The next possible source of error was the incorrect input of fin dimensions into the model, but all input values matched the provided geometrical inputs. The remaining possibility of error was a misunderstanding of what physical parameters were related to specific variables for the fin geometry. The possibility of using the wrong cross-sectional area of the fin (confusing L with H) was evaluated but these variables were used correctly. The preliminary model mixing up wetted perimeter with regular fin perimeter or confusing hydraulic diameter with the physical gap between fins was also possible. None of these common mix-ups were present in the preliminary model upon a triple check. This led to the conclusion that there may have been some errors within the experimental data provided.

One revision was made to the initial model regarding the Thermal Interface Material (TIM). Feedback provided on the Interim Report noted that the TIM thickness was too high, which prompted a recalculation. The necessary TIM resistance was calculated as the difference between the ideal resistance of the cold plate and the resistance of both the cold plate and the base, as shown below.

$$R_{TIM} = R_{Ideal} - R_{Cold Plate} - R_{Base} \quad (1)$$

The necessary resistance for the TIM was calculated to be $1.075 * 10^{-5} \text{ }^{\circ}\text{C/W}$. Using the chosen TIM thermal conductivity of $8 \text{ W/m}^{\circ}\text{C}$, the necessary thickness was calculated using the conduction resistance equation, accounting for the thickness (t), the thermal conductivity (k), and the cross-sectional area (A).

$$R = \frac{L}{k * A} \quad (2)$$

The calculated TIM thickness was 0.08 mm, as opposed to the value of 1.0 mm presented in the Interim Report.

1.2 Development of Pin Fin Cold Plate Design

The pin fin cold plate was developed using the same the derivation as the parallel pin fin cold plate in the Interim Report. The goal of the design was to achieve a total thermal resistance of about 0.02 °C/W, the ideal thermal resistance as derived in the Interim Report. There were several key differences in the equations used, specifically the Nusselt correlation and the pressure drop calculation.

Equation 3 calculated the Reynolds number (Re) as a function of the hydraulic diameter of the fin array (D_H), the kinematic viscosity of the water (ν), and the internal velocity of the water through the cold plate (U_W).

$$Re = \frac{U_W * D_H}{\nu} \quad (3)$$

Equation 4 calculated the Nusselt correlation (Nu) as a function of the Reynolds number (Re) and the Prandtl number (Pr) of water. The correlation used was valid for Reynolds numbers between 0-800, and the Reynolds number in this design never exceeded 500.¹ This differed from the Nusselt correlation used in the Interim Report, which was assumed to be a constant 7.54. This was the asymptotic value for fully developed laminar flow through a channel, but this cold plate was not designed with channels.

$$Nu = 0.143 * Re^{0.615} * Pr^{\frac{1}{3}} \quad (4)$$

Equation 5 calculated the convective heat transfer coefficient (h) as a function of the Nusselt correlation (Nu), the thermal conductivity of water (k_w), and the hydraulic diameter (D_H).

$$h = \frac{Nu * k_w}{D_H} \quad (5)$$

Equation 6 calculated the thermal resistance of the pin fin array ($R_{fin\ array}$) as a function of the mass flow rate of water in the system ($\dot{m}$), the specific heat of water at a given temperature (c_p), the pin fin array efficiency (η_0), the convective heat transfer coefficient (h), and the total area of the pin fin array (A_t). A_t and η_0 were calculated using the same equations as the parallel fin array in the Interim Report.

$$R_{fin\ array} = \frac{1}{\dot{m} * c_p (1 - \exp\left(\frac{-\eta_0 * h * A_t}{\dot{m} * c_p}\right))} \quad (6)$$

Equation 7 calculated the Darcy friction factor (f_D) as a function of the Reynolds number (Re). This calculation of f_D was valid for the pin fin cold plate because the maximum Reynolds number simulated by the model was about 450 (well within the bounds listed for this equation,

which was around 200-2000).² This equation accounted for the gaps between the pins, increasing the pressure drop as compared to a channel flow design.

$$f_D = 1.2 * Re^{-0.2} \quad (7)$$

Equation 8 calculated the major pressure loss through the pin fin array (ΔP_{Major}) as a function of the combined length of the fins (L), the Darcy friction factor (f_D), the hydraulic diameter (D_H), the density of the water (ρ), and the velocity of the water squared (U_W^2).

$$\Delta P_{Major} = (f_D) \frac{L}{D_H} \frac{\rho U_W^2}{2} \quad (8)$$

The minor pressure losses due to the inlet and outlet manifolds (ΔP_C) were calculated by Equation 9 as a function of the minor loss coefficient (K_C), the density of the water (ρ), and the velocity of the water squared (U_W^2).

$$\Delta P_C = K_C \frac{\rho U_W^2}{2} \quad (9)$$

The change in minor pressure was negative for the inlet, and positive for the exit, and since the K_C value was the same, and the density and velocity remained unchanged, they canceled each other out. Therefore, the only notable pressure drop through the cold plate was due to the frictional losses on the pins. The pin fin cold plate geometry and results are listed in Table 1. For the best comparison between the two plates, the pin fin cold plate was also designed to achieve optimized thermal resistance at 1.2 LPM.

Table 1. Pin Fin Geometry

Parameter	Value
Total Thermal Resistance	.0199 °C/W
TIM Resistance	0.0025 °C/W
Cold Plate Thermal Resistance	0.01638 °C/W
TIM Thickness	0.05 mm
TIM Thermal Conductivity	8 W/m°C
Fin Height	3 mm
Fin Thickness	0.5 mm
Fin Spacing	0.4 mm (uniform grid)
Total Array Length	50 mm
Base Thickness	1 mm
Base Resistance	0.001309 °C/W
Fin Array Efficiency	72.94%
Number of Fins	3136 (56x56)
Shape of Fins	Square Pins
NTU	0.598

Volumetric Flow Rate	1.2 LPM
Total Pressure Drop	0.4334 PSI

1.3 Comparison Between Parallel Fin and Pin Fin Designs

Comparisons were made between the thermal and hydraulic performance of the original fin shape (parallel rectangular channels) and the pin fin design. For low flow rates (under 1.5 LPM), as seen in Figure 1, the parallel fin design provided a lower thermal resistance than the pin fin design. For example, at a flow rate input of 0.4 LPM, the pin fins yielded a thermal resistance of 0.033 °C/W, a 13% increase over the parallel fin thermal resistance of 0.029 °C/W. When observing higher flow rate values, such as 1.2 LPM (the flow rate optimized for the selected cold plate fin dimensions), the thermal resistance difference between the pin and parallel fins began to converge. At the highest observed flow rates, both the pin fins and parallel fins provided adequate thermal resistance values below the target thermal resistance.

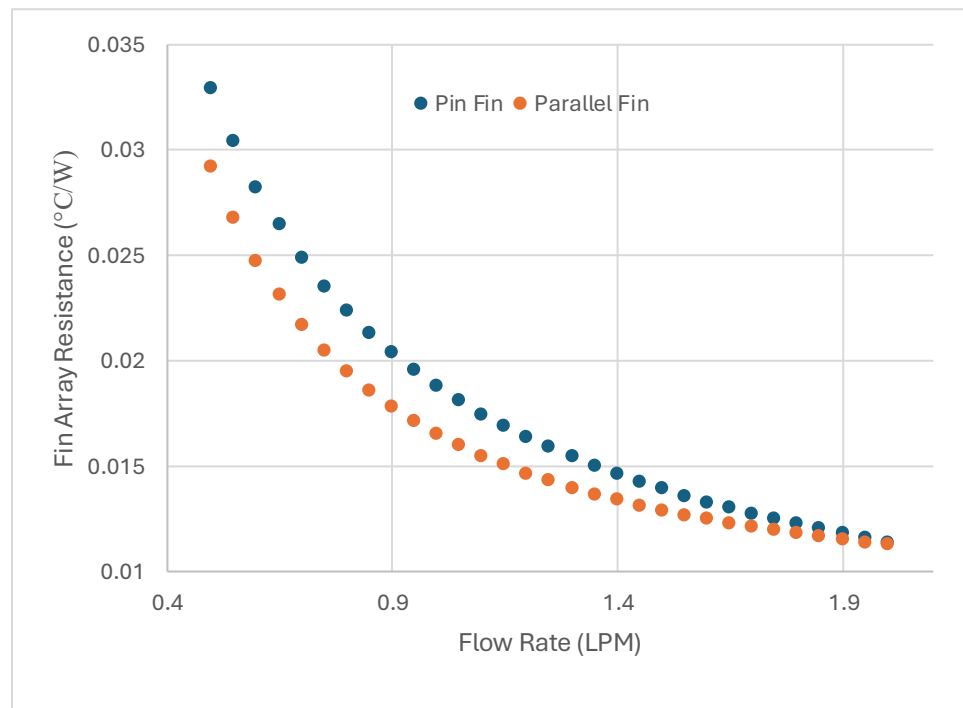


Figure 1. Fin Array Resistance vs Flow Rate by Fin Type

The deciding factor to use a pin fin or parallel fin design on the flow loop cold plates was the pressure drop. At all observed flow rates, the pin fins created a higher pressure drop in the system than the parallel fins. This difference in pressure drop between fin designs increased with flow rate, as shown in Figure 2. At the flow rate optimized for the cold plate (1.2 LPM), the pressure drop in the pin fin cold plate was about 0.47 psi. This was more than three times the pressure drop created by the parallel fin cold plate at the same flow rate (0.13 psi).

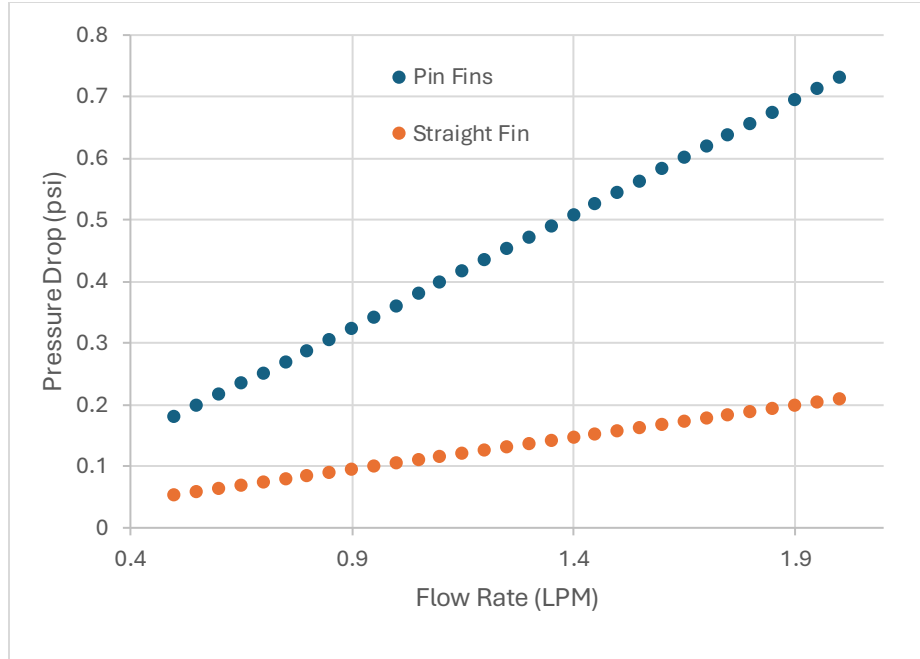


Figure 2. Pressure Drop vs Flow Rate by Fin Type

2. Server Flow Loop Hydraulic and Thermal Design

The hydraulic server flow loop was designed using two different diameter copper pipes, as well as a combination of ball valves, dripless quick disconnects, eight cold plates, and three custom manifolds. Each manifold integrated fittings into a single part, and the system complexity was decreased due to a reduction in possible points of pressure drop. The manifolds eliminated the need for regular maintenance by taking the place of three crosses, seven tees, and four elbows that would have been used to divert and reconnect the flow. The 14-inch-long central manifold contained eight circular cross-section protrusions with a 0.5-inch diameter (four on each opposing side) to split the incoming cold water into eight separate branches of parallel flow. The minor pressure loss coefficients for the various fittings were still used to calculate pressure drop. These coefficients were smaller for the same fittings to account for the manifold's elimination of pressure drop throughout its length due to the isobaric assumption explained in Section 3 in greater detail.

The center manifold was designed to be 14 inches long. This length allowed for each of the GPUs to be separated 2 inches apart from each other while still fitting within the maximum server length of 16 inches. The outer two manifolds were 16 inches long, taking up the entire possible server length to allow for the outlet flow of each parallel branch to reconnect at a single outlet pipe. Only one dripless quick disconnect fitting was added to each branch of the server loop, rather than adding two fittings (one on either side of the cold plate). The second dripless connection was added outside of the main cold plate branch loops to both groups of the four parallel branches (left and right sides of the center manifold). Shifting the location of the second quick connection allowed the loop layout of components to fit widthwise within the 16-inch

maximum width. Each dripless quick disconnect was 2 inches in length, and each cold plate was also about 2 inches in length. The manifolds were all 1.5 inches in diameter, so the total width of the two branches' components in parallel was 12.5 inches. The main factors considered when determining the flow loop design were the pressure drop through the loop, the total flow rate, and the functionality of the cold plates. The optimal cold plate flow rate determined in the Interim Report was 1.2 LPM, so the goal of this report was to maintain that flow rate to ensure effective cooling of the chips. A schematic of the designed flow loop is shown below in Figure 3.

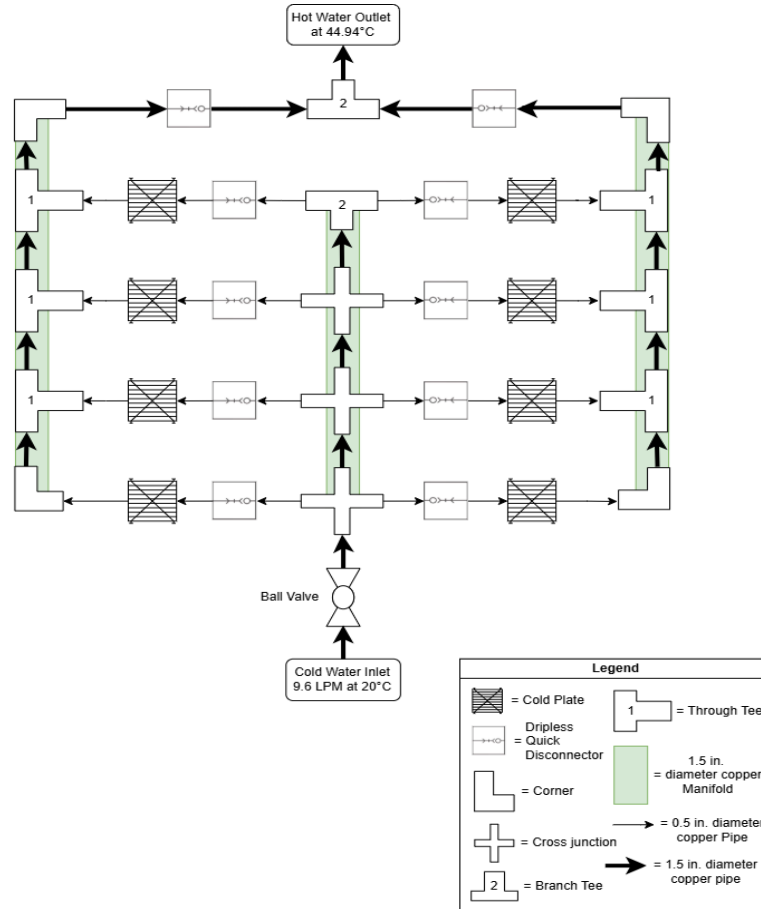


Figure 3. P&ID Diagram of Server Flow Loop

The pressure drop through the flow loop was due to both major and minor head losses. The total flow rate affected the friction of the flow in the pipes, resulting in major head loss (P_M) and affected pressure drop through the chosen fittings, resulting in minor head loss (P_m). The major head loss, or pipe friction, was calculated by Equation 10 as a function of the friction factor (f), the speed of the fluid (U_w), the density of the cooling fluid (ρ), and a ratio of the pipe length (L) to diameter (D).

$$\Delta P_M = f \left(\frac{L}{D} \right) \left(\frac{\rho U_w^2}{2} \right) \quad (10)$$

The pressure losses (minor head loss) were due to the fittings and components throughout the flow loop. The calculation for the minor loss was a function of the loss coefficient (K), a factor dependent on the type and diameter of the component. The K values for the components used in the flow loop are shown below in Table 2. The friction factors for the straight pipe were found using the Moody Chart [A-1], based on the calculated Reynolds number through each pipe.

Table 2. Loss Coefficients for Components in Flow Loop

Component	Friction Value
Manifold Branch ³	$K = 1.26$
Branch Tee ³	$K = 1.26$
Elbow (90°) ⁴	$K = 0.63$
Ball Valve ³	$K = 0.05$
Dripless Quick Disconnect ⁵	$K = 2.5$
Straight Copper Pipe (0.5 in)	$f = 0.05$
Straight Copper Pipe (1.5 in, 4.8 LPM)	$f = 0.042$
Straight Copper Pipe (1.5 in, 9.6 LPM)	$f = 0.038$

These loss coefficients were used in the minor head loss calculation, which also accounted for the density of the fluid and the velocity. The equation for the minor head loss due to the components in the flow loop was:

$$\Delta P_m = K \left(\frac{\rho U_w^2}{2} \right) \quad (11)$$

The velocity of the fluid was directly related to the flow rate of the system. The pressure loss equations above demonstrate a proportional relationship between the velocity and pressure drop, meaning that as the flow rate increased, the pressure drop increased. To determine the exact values in this relationship, the total pressure drop through the system was calculated for a series of flow rates. This was the total of the major head loss, the minor head loss, and the pressure drop through the parallel fin cold plate, which was calculated to be 0.124 psi in the Interim Report. The effects on the pressure drop due to a varied flow rate are shown below in Figure 4.

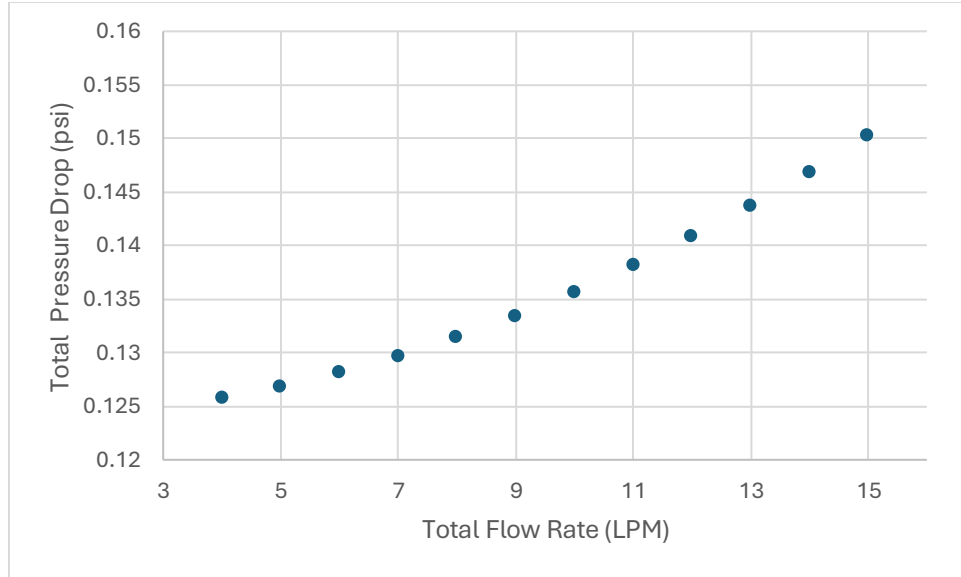


Figure 4. Flow Loop Total Pressure Drop vs Flow Rate

The lowest flow rate would have resulted in the lowest pressure drop, but the system still required a flow rate large enough to effectively cool the chips. Since the flow loop design utilized eight branches, each supplying one cold plate, the total flow rate was determined to be eight times the optimal flow rate through one plate. The optimal total flow rate was 9.6 LPM, which resulted in 1.2 LPM being distributed to each plate. The pressure losses at this flow rate were shown in Table 3. The pressure drop due to each type of component (excluding the cold plate) was displayed in Figure 4.

Table 3. Pressure Losses Through the Flow Loop

Pressure Type	Pressure Loss (psi)
Minor Head Loss	0.00977
Major Head Loss	0.00104
Cold Plate Pressure Loss	0.1239
Total Pressure Loss	0.1247

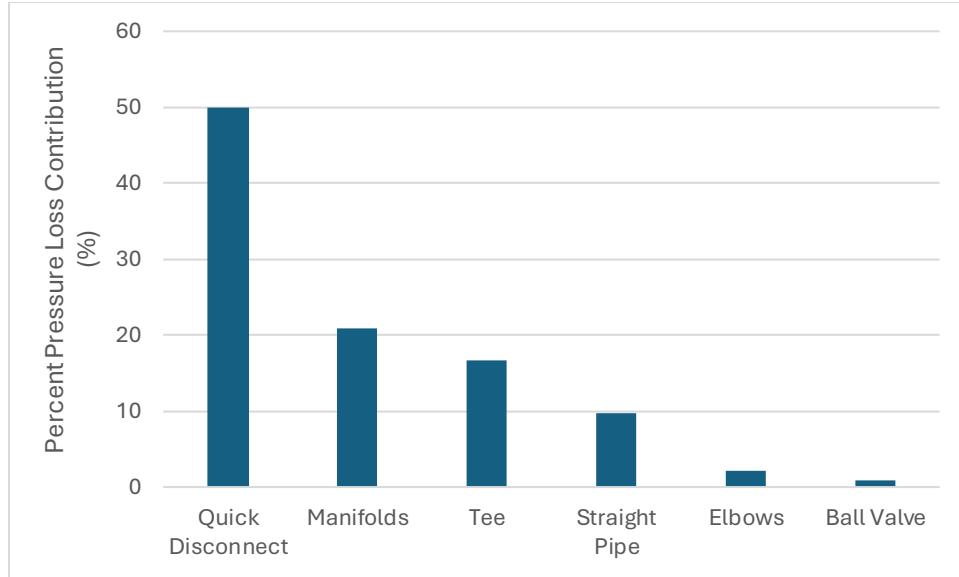


Figure 4. Pressure Loss Through Different Components

The case temperature of each processor chip in the flow loop was predicted using the model from the Interim Report. Without temperature variance of inlet water temperature through the cold plates, the calculations of the GPU case temperature were the same for all eight processors and proved that the maximum case temperature would not exceed the maximum design constraint (60°C).

The temperature of the cold plate was calculated as a function of the total thermal resistance of the cold plate (R_T), the heat generated by the chip (q), and the inlet water temperature (T_i) in Equation 12. This equation was the thermal version of Ohm's law, rearranged to solve for the chip temperature.

$$T_C = q * R_T + T_i \quad (12)$$

The chip temperature was calculated as 59.44°C. This was right below the target temperature of 60°C. Keeping the temperature close to the desired temperature was allowed for maximum performance in the processor without cooling the chip excessively, which would have created more pressure losses as discussed in the Interim Report. The chosen flow loop design of eight parallel loops allowed for greater control and a lower margin of error when controlling the chip temperature, since the water inlet temperature was constant for every chip.

3. Conclusions and Recommendations

The initial flow loop design used two parallel branches. Each branch contained four cold plates in series with dripless quick disconnects in between for ease of maintenance. This layout allowed for a more compact design (system fits into the maximum length and width constraints with more wiggle room). This layout was deemed ineffective due to its failure to maintain system thermal efficiency, which was caused by the same water temperature concerns (series plates receive inlet water at too high a temperature to achieve the desired maximum thermal

resistance) outlined above. This dual-branch design also created an increase in the pressure drop experienced by the fluid in the system. There were four cold plates and five dripless quick disconnectors (the two largest contributors to pressure drop, as indicated in Figure 4) in series on each branch of the dual-branch design. This design caused the pressure drop of each successive component to sum to a larger total pressure drop. For these reasons, this initial layout was no longer considered.

The failure of the dual-branch flow loop inspired the entire server loop (all eight cold plates) to be placed in parallel for two main reasons. First, organizing the cold plates in parallel allowed for a maintained system thermal efficiency and minimized pressure drop. Second, maintaining thermal efficiency was necessary to ensure the thermal resistance simulation calculated in the Interim Report remained valid. The maximum calculated total thermal resistance of the cold plate was $0.02\text{ }^{\circ}\text{C/W}$, keeping the chip from overheating to a temperature above 60°C . The optimized cold plate geometry yielded a total thermal resistance calculation of $0.016\text{ }^{\circ}\text{C/W}$, nearing the maximum temperature constraint. This $0.016\text{ }^{\circ}\text{C/W}$ thermal resistance calculation depended significantly on the temperature of the inlet water (chosen to be 20°C) being fixed. The calculated water temperature at the exit of one cold plate was 43.94°C . The temperature after each cold plate in series is shown below in Table 4. Even if the server loop had four parallel branches with two cold plates in series in each, the incoming water temperature for the second plate in series would be the outlet temperature of the first plate (43.94°C). This would cause the required thermal resistance of the second plate in series to be far below the designed $0.02\text{ }^{\circ}\text{C/W}$ resistance, which resulted in every chip after the first one in the series to overheat. The water temperature that entered the third and fourth chips would have been higher than the chip temperature, having a heating effect instead of a cooling effect.

Table 4. Outlet Water Temperature of Cold Plates in Series

Cold Plate	Outlet Water Temperature ($^{\circ}\text{C}$)
1	43.94
2	67.86
3	91.79

The server loop containing eight parallel branches also worked to minimize the total pressure drop. Pressure drop due to individual system components was only summed when the components were oriented in series. Two cold plates in series yielded double the pressure drop of a single cold plate, but two cold plates in parallel yielded the pressure drop of a single cold plate (similar to how the total resistance of two resistors in series equals the resistance of each component). Since all of the cold plates in the loop are organized along independent parallel branches, the pressure drops from the cold plates did not have to be added. The total pressure drop of the system was calculated by finding the pressure drop through one of the branches. The pressure drop within one branch consisted of only one cold plate, tee, cross, and dripless quick

disconnect. This allowed for the total loop system pressure drop to be multiple times smaller than it would have been if there had been fewer branches with more components in the series.

The manifolds were designed and implemented into the flow loop with a circular cross-section 1.5 inches in diameter. Each manifold had protrusions coming out of either one or both sides, depending on the manifold location. These protrusions acted as a combination of various fittings (through and branch tees, elbows, and crosses). Manifolds can be assumed internally isobaric when the length-to-diameter ratio is less than 50.⁶ The 14-inch length and 1.5-inch diameter of the designed manifolds allowed for a ratio of 28:1 length-to-diameter ratio. The validated isobaric assumption contributed to the integration of fittings into a manifold, causing a slight reduction in pressure drop. Posted minor pressure loss coefficients for tees and crosses included two aspects (pressure loss through and branched). This created unnecessary pressure loss of water through the crosses as it was working its way to the furthest two cold plates in parallel. The manifold's isobaric assumption holds that there is no internal pressure loss of the fluid through the manifold. This elimination of pressure loss through the majority of the fittings in the system informed a decision to utilize manifolds, as the constant pressure throughout the manifold nearly guaranteed that the flow rate that was diverted into each of the branches would be the same (1.2 LPM).

The flow loop and cold plate design resulted in a complete liquid cooling design for a singular server rack. The design considered pressure drop, the effectiveness of the cold plates, the ease of maintenance on the system, and ultimately resulted in a very efficient and low pressure drop design.

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7. ChatGPT was used to find heat transfer equations for pin fins, such as the Nusselt correlation. The equations' validity were confirmed by prompting ChatGPT for the exact source it used to pull information from the internet. ChatGPT was also used to brainstorm possible server flow loop branch combinations to minimize pressure drop.

Appendix

A. Moody Chart for Pipe Friction Factors

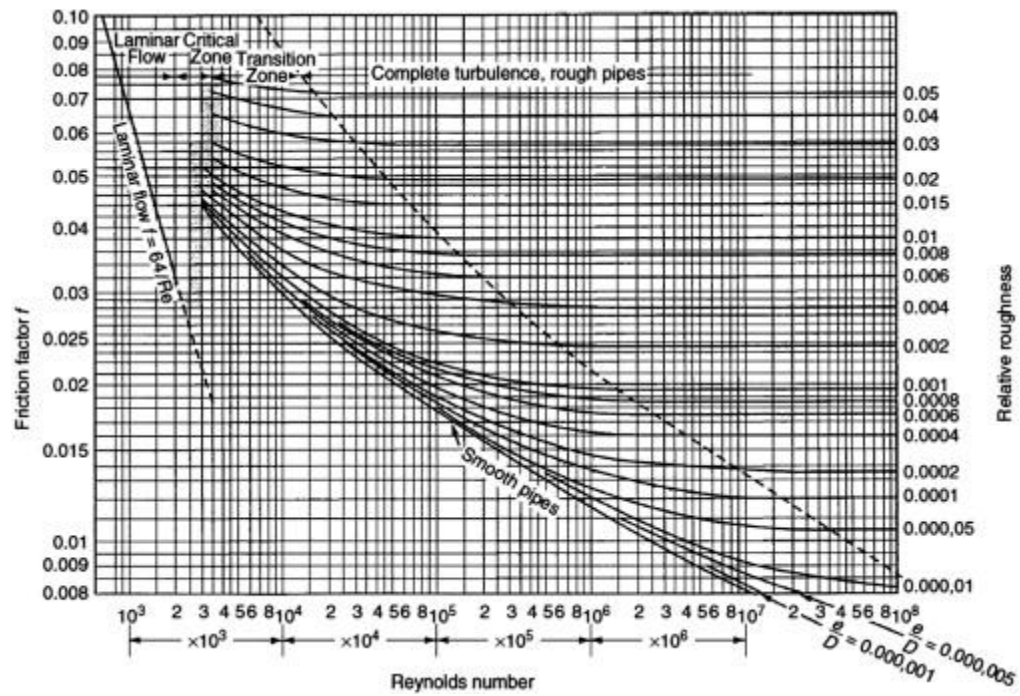


Figure A-1. Moody Chart